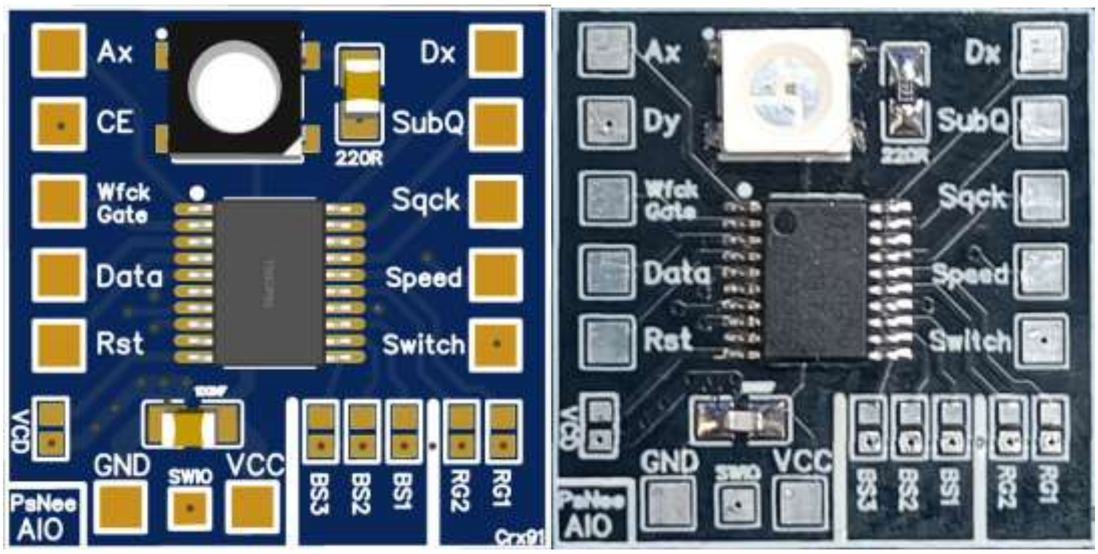
PsNee_Aio: the ultimate mod for Ps1!



The ultimate mod based on PsNee, my OneNee and my latest Jap_Bios_Unlocker.
Unleash the power of your Ps1!

Coded for the "10 cent" MCU CH32V003 using the powerful ch32fun development environment, PSNee_Aio is the ultimate mod for Playstation 1 console!

Features:

- Fully compatible and stealth with all PS1 models and BIOS!
- Unlock JAP bios, allowing boot of PAL and USA games;
- Unlock PAL PSone (SCPH-102) bios, allowing boot of USA and JAP games;
- Allows boot backup games and backup VCD on SCPH-5903 consoles;
- It uses an alternate and universal method of JAP bios patching “on the fly”. No more bios crash, no more timing problems and no more need to recompile the code for each Jap bios version;
- Jumper settable, so you can move the chip from various models or change the settings\features without recompiling the code again!;
- Precompiled BIN image ready to flash on ch32v00f4p6, no need to mess with Arduino IDE and compilers!;
- Ws2812b RGB output led to know the status of the chip!;
- Fully open-source. (If you see this mod sold online, I am not getting a dime!);
- Easily upgradeable onboard using a single wire (SDI) through the SWIO pin;
- Original PsNee code by ramapcsx2 with kalymos updates and optimizations;
- VCD detection code made by RepairBox;
- Jap_Bios_Unlocker code made by Crx91 (me).

Supported MCU:

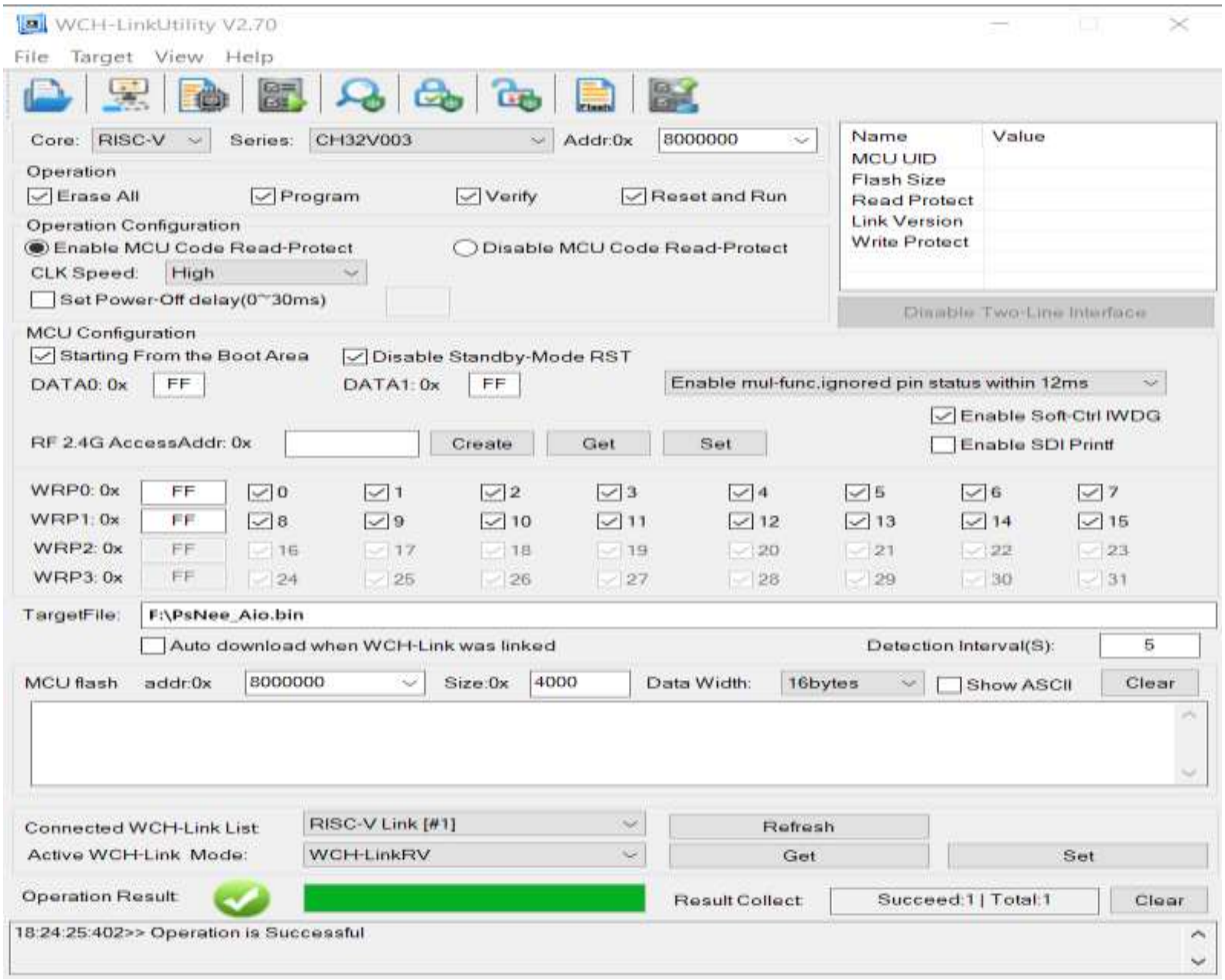
- Ch32v003f4p6
- Ch32v003f4u6 (PCB and schematics not available at the moment)

Prerequisites:

- WCH-LinkE programmer (pay attention to the E).
- WCH LinkUtility software.

Step 1 - Programming the MCU:

- Download [this]() repository.
- Install the WCH LinkUtility software and the relative drivers (all the the needed software/files are zipped in the "tool" directory).
- Connect the LinkE programmer and check if is in RISCV mode (Blue LED should be always off when idle).
- If not, disconnect and reconnect the programmer holding down the "Mode-S" button.
- Open LinkUtility and check if the tool sees the programmer.
- Set the core (RISC-V) and the series (CH32V003).
- Click on the folder icon (or press ALT+F1)
- Select the BIN compiled file (you can find the bin file in the [BIN]() folder).
- Connect the chip to the programmer.
- Program the chip via Target -> Program (or press F10). Program the chip via Target -> Program (or press F10).



- Now you have your PsNee_Aio on WCH CH32V003 chip!
- For the installation, follow the settings instructions above.

Step 2 - Console region identify:

You can identify your console region looking at the rear of the unit, you will find a sticker.
Here you can read the console model starting with SCPH-xxxx:

SCPH-xxxx Model no:	Region	Note
1000, 3500, 5000, 5500, 7000, 7500, 9000, 100	JAP	To run USA and PAL backup you need set the “JAP_Bios_Unlock” fix.
1001, 5001, 5501, 7001, 7501, 9001, 101	USA	
1002, 5002, 5502, 7002, 7502, 9002	PAL	
102	PAL	To run JAP and USA backup you need to set the “OneChip” bios fix.
5903	JAP	Recommended to set the “VCD feature”.
5503, 7003, 7503, 9003, 103	JAP	

Or you can search, on the motherboards from PU-8 [Late] to PM-41, for a 52-pin square chip on. This is the mechacon; look for the first row numbers. The two digits after the C identify the injection region required:

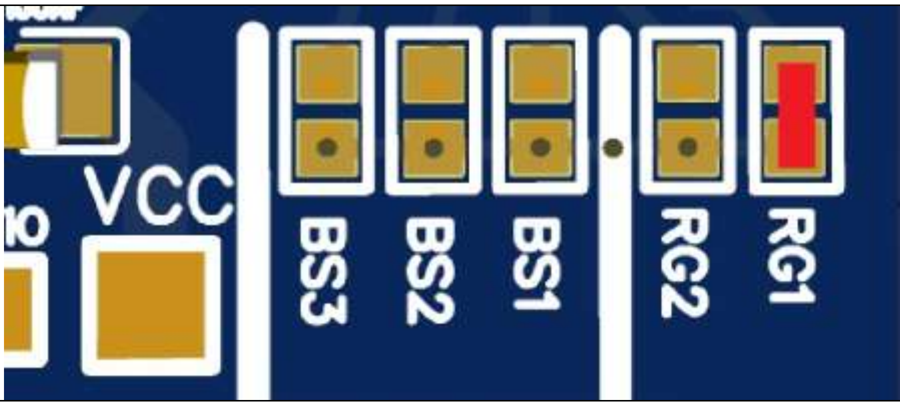
<table>
<tr><td rowspan="4">
</td><td>Code</td><td>Region</td></tr>
<tr><td>C10xx</td><td>JAP</td></tr>
<tr><td>C20xx</td><td>USA</td></tr>
<tr><td>C30xx</td><td>PAL</td></tr>
</table>

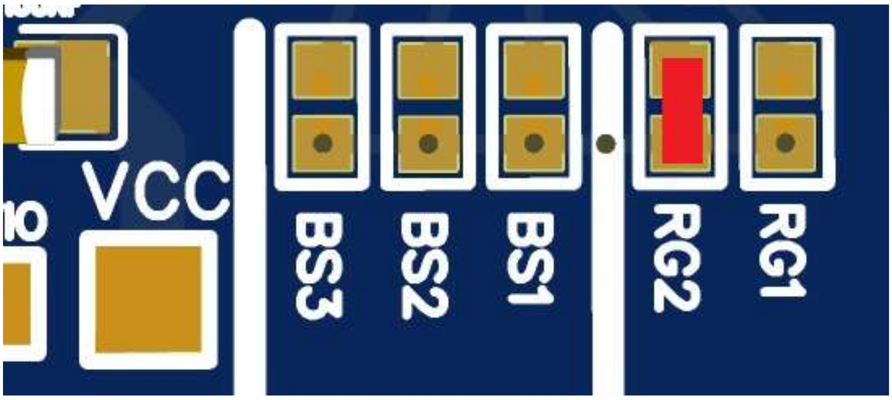
NOTE about additional settings:

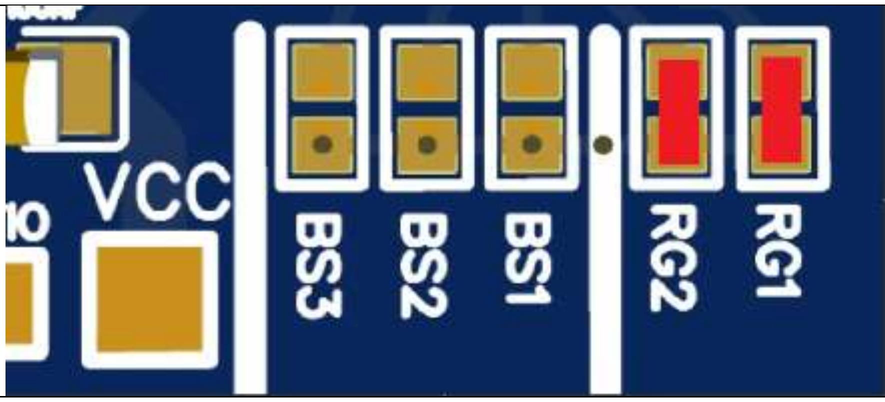
- **All JAP BIOSes:** To play USA and PAL games you will need to set the “JAP_Bios_Unlock” and solder the extra required points labeled *reset* (only on fat models), *speed*, *Bios_Dx* and *Bios_CE*;
- **Only PAL PsOne SCPH-102 BIOS:** To play JAP and USA games you will need to set the “OneChip” bios fix and solder the extra required points labeled #Bios_Ax# and *Bios_Dx*;
- **For SCPH-5903:** Is recommended to set the VCD feature.

Step 3 - Jumper settings:

- Injection region set:

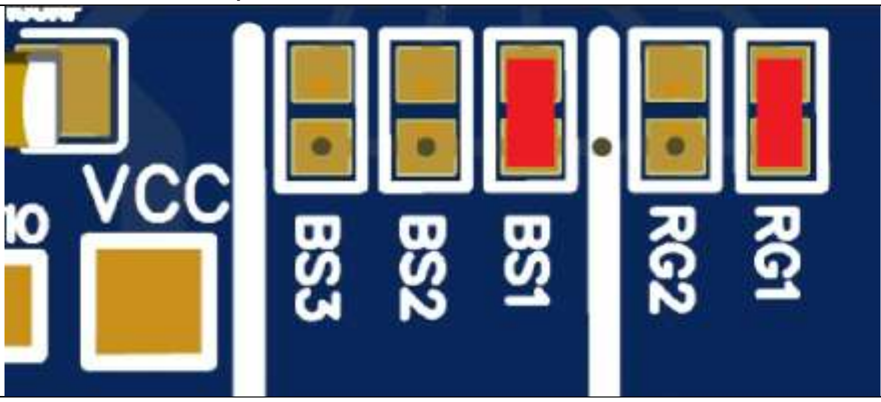
For **PAL** consoles	
Bridge RG1 only (or put PC5 to GND)	
NOTE: **SCPH-102 (PM-41 v1\v2)** to boot USA and JAP games, you need to set the OneChip Bios fix. Read the “Bios unlock setting” section above for the details.	

For **USA** consoles	
Bridge RG2 only (or put PC6 to GND)	

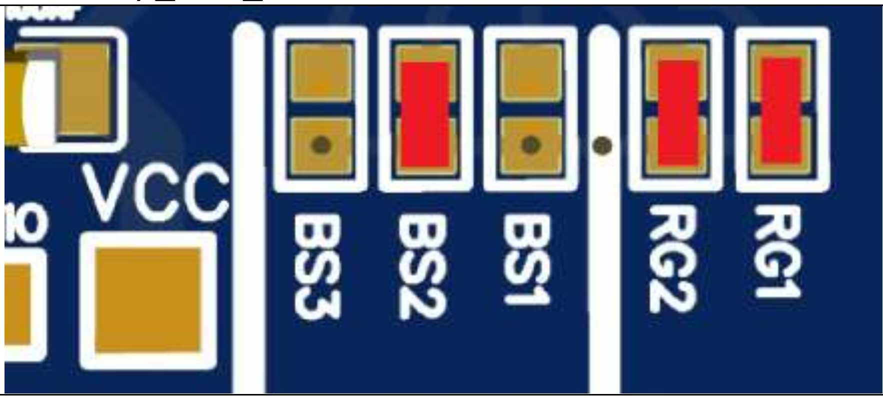
For **JAP** consoles	
Bridge RG1 and RG2 (or put PC5 and PC6 to GND)	
NOTE: On all **JAP BIOSes** to boot USA and PAL games, you need to set the “Jap_Bios_unlocker” fix. Read the “Bios unlock settings” section above for the details.	

- Bios unlock settings:

For the **PAL PsOne SCPH-102 with PM-41 v1\v2** motherboards the BIOS allows only booting of PAL games. To unlock other regions boot you need to enable the “OneChip BIOS fix”:

<table>
<tr><td colspan="2">OneChip BIOS Fix set</td></tr>
<tr><td>Bridge BS1 only
(or put PC3 to GND)</td><td>
</td></tr>
<tr><td colspan="2">Requires soldering additional *points* labeled: *Bios_Dy*, #Bios_Ax# on the diagrams!</td></tr>
</table>

For the consoles with **JAP BIOSes**, you need to enable the “Jap_Bios_Unlocker” feature to allow booting of other region games:

<table>
<tr><td colspan="2">Jap_Bios_Unlocker set</td></tr>
<tr><td>Bridge BS2 only
(or put PC2 to GND)</td><td>
</td></tr>
<tr><td colspan="2">Requires soldering additional *points* labeled: *Bios_CE*, *Bios_Dx*, *Speed* and *Reset* (only on fat models) on the diagrams!</td></tr>
</table>

Enable\disable BIOS fix via external switch feature:

Optionally, you can enable\disable the set BIOS fix, by connecting an external switch to the modchip PCB pad labeled “Switch” and a GND point on the console motherboard.
(If you are not using the PCB, the "switch" pad is equivalent to the PC4 pin on the ch32v003).
- Closing the switch before the boot of the console (or before a reset), the BIOS fix will be disabled.
- Opening the switch before the boot of the console (or before reset), the set BIOS fix will be re-enabled.

VCD feature:

Only on VCD SCPH-5903 (PU-16) you must enable the VCD feature. This mode filters and differentiates SUBQ data streams from game discs and VCD discs.

VCD feature enable	
Bridge the VCD jumper on the modchip PCB (or put PC0 to GND).	

Step 4 - Modchip installation:

After you have set the chip with appropriate jumpers, follow the wiring diagrams in the "Install" folder based on your console motherboard.
- Pay attention to the additional required labeled ***points*** if you are using any BIOS fix!
- Always power the chip with Psx 3.3vdc \ 3.5vdc power.
- If you don't know which revision of the motherboard you have, search for a label like PU-xx or PM-xx written on the board itself or consult this wiki:
https://www.psdevwiki.com/ps1/Motherboards

WS2812b output meaning:

The WS2812b shows different colors based on the setting made or issue detected:

WS2812b color	Meaning
White	Detection of which method of injection to use (old based on gate mode or newer based on wfck mode).
Magenta	Applyng the PAL PM-41v1\v2 Onechip BIOS fix.
Light blue	Applyng Jap_Unlocker_Bios fix.
Green	Injecting PAL string.

Orange	Injecting USA string.
Blue	Injecting JAP string.
Red	Invalid BIOS fix set, check BS1\BS2\BS3 jumpers.
Blinking Red	Injection region not set, check RG1\RG2 jumpers.

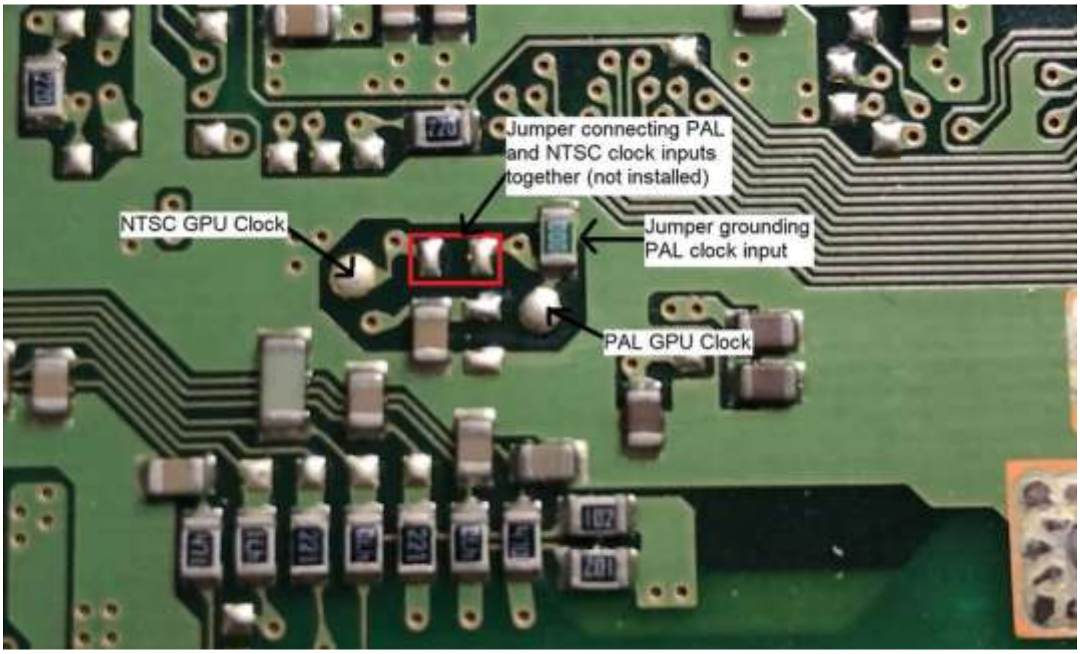
Video mode issues:

If you use a composite cable on a motherboard from PU-7 to PU-18 and run PAL software on NTSC units (or viceversa) you can have video issues, like Black and white image, rainbow effect or no video at all.
On all motherboard revisions you will always have a framerate speed error of about 1% (slower if run NTSC on PAL, faster if you run NTSC on PAL)
Unfortunately that kind of issue can't be fixed with a modchip, but you can fix or mitigate the problem various methods:

1. Using an RGB cable (but you will always have the gameplay speed issue);
2. For PU-7 to PU-18: Install a secondary osc on the specific GPU clock pin (NTSC uses 53.693(18) Mhz and PAL uses 53.203425 Mhz);
3. For PU-20 to PM-41: Force video output with a link on two points (search for the fix on the web, the gameplay speed issue remains);
4. Installing a DFO and fix all the issues.

On very early NTSC PU-7 and PU-8 board when switched to PAL mode you can have a black screen. This is caused by a lack of the PAL clock input on the GPU, resulting in it freezing when switched to PAL mode.

You can try to fix this by removing the green jumper currently grounding the PAL clock and moving it to the middle position marked with a red box - but this only works with later GPU revisions. If you find you get no video at all, just move it back. (Thanks to Trimesh for the fix)



PINOUT for the WCH CH32V003f4p6:

```
                   ___ ___
   Bios_Ax (PD4) -|   U   |- (PD3) Bios_Dx
 WFCK_GATE (PD5) -|       |- (PD2) SUBQ
      DATA (PD6) -|       |- (PD1) [SWDIO]
   [Reset] (PD7) -|       |- (PC7) SQCK
   Bios_CE (PA1) -|       |- (PC6) reg_Bit0
     Speed (PA2) -|       |- (PC5) reg_Bit1
             GND -|       |- (PC4) Switch
   WS2812B (PD0) -|       |- (PC3) bios_Bit0
             VCC -|       |- (PC2) bios_Bit1
  VCD_mode (PC0) -|       |- (PC1) bios_Bit2
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```

Special Thanks to:

ramapcsx2, kalymos, SpenceKonde, oldcrow, mayumi, arduino community, ch32fun community, Infrid and lots of people that can't remeber now.

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